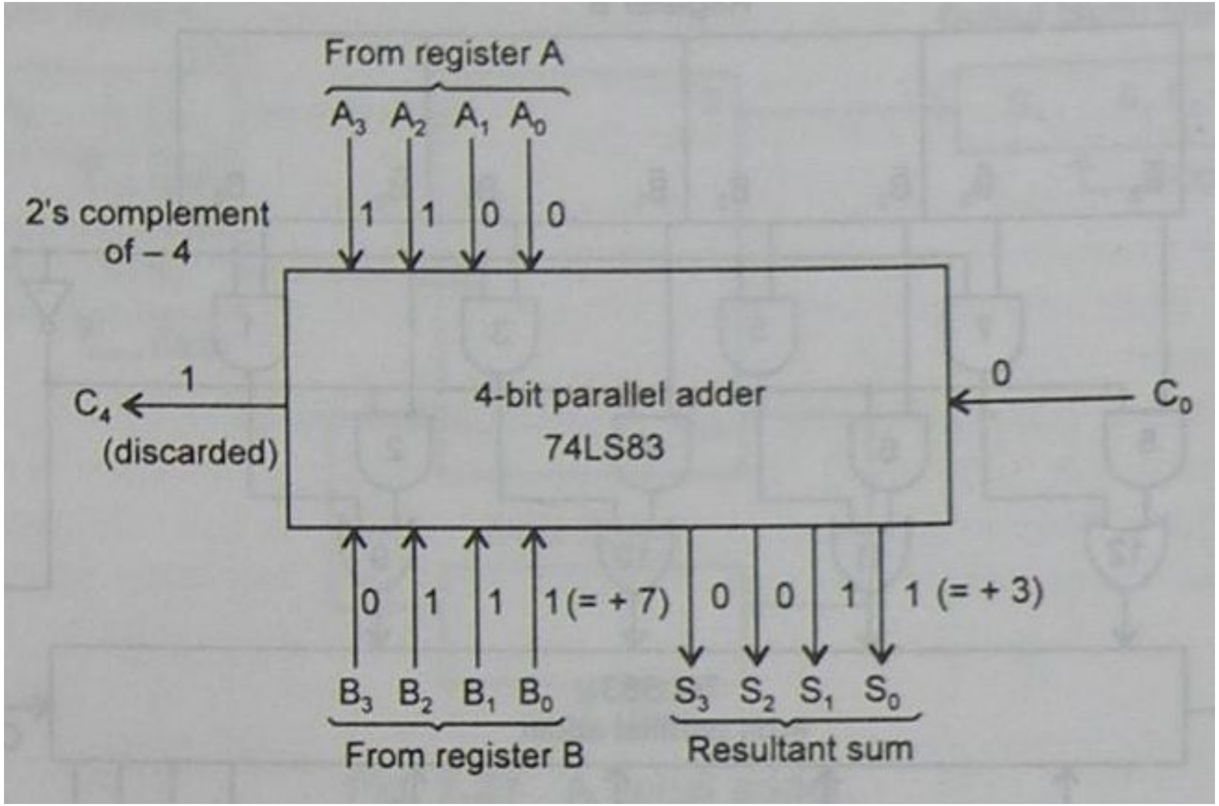
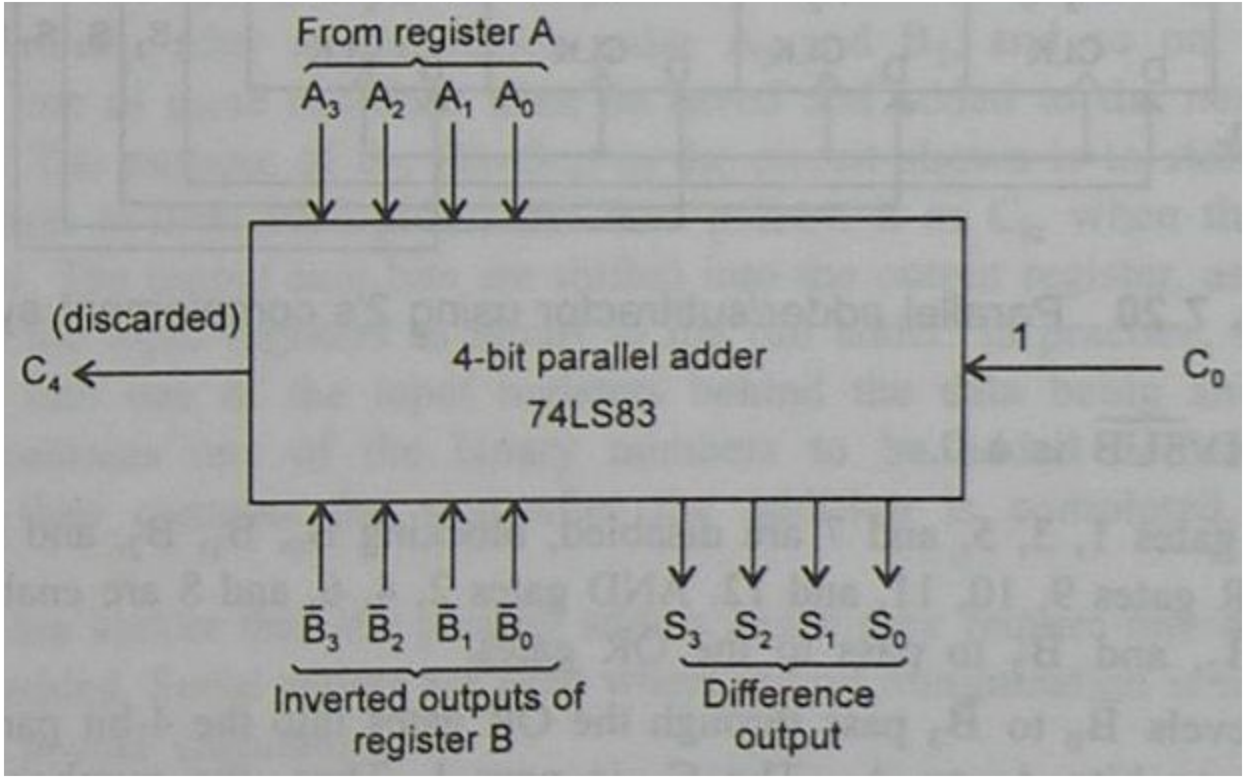
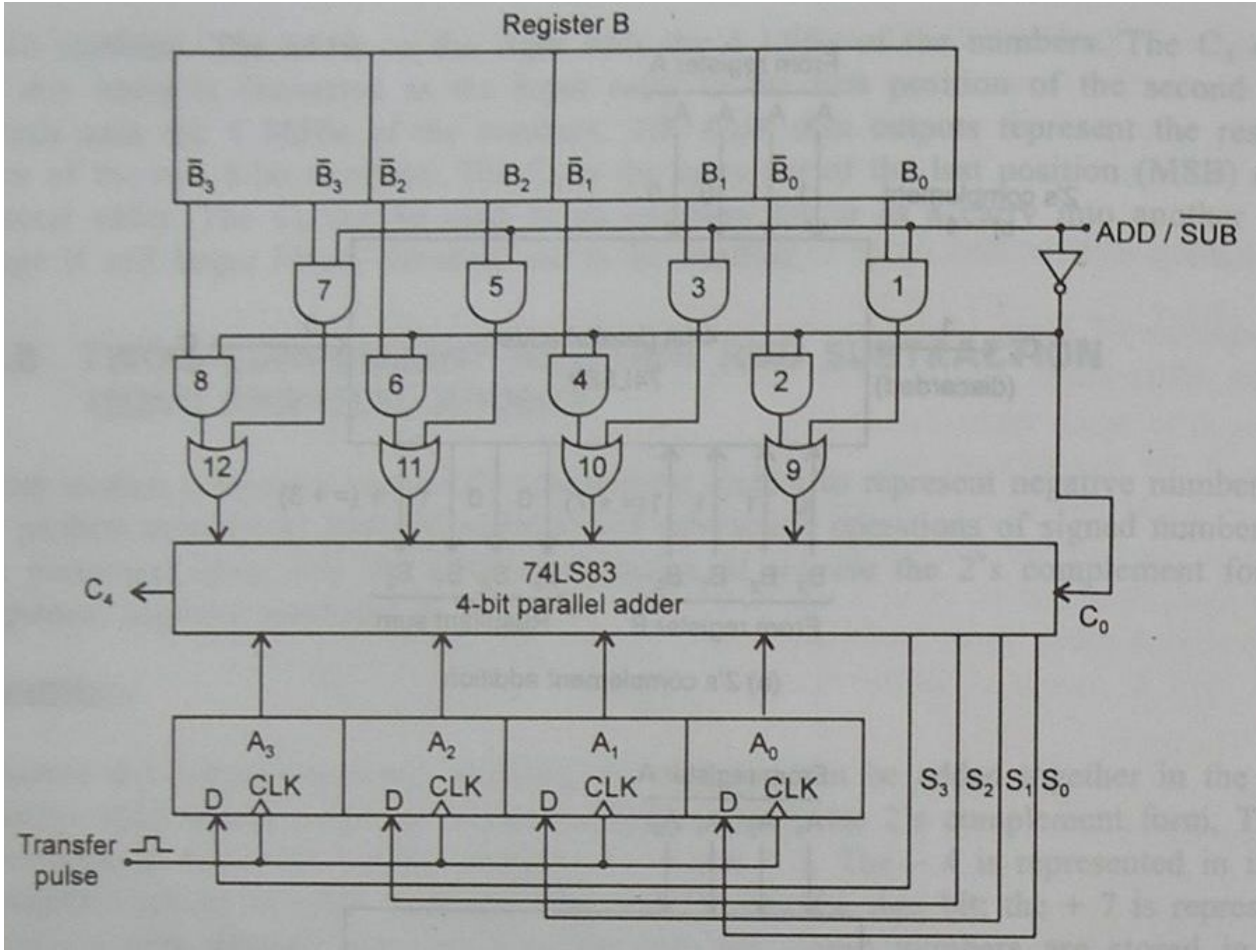
Adders and Subtractors

2’s Complement Adder



2’s Complement Subtractor





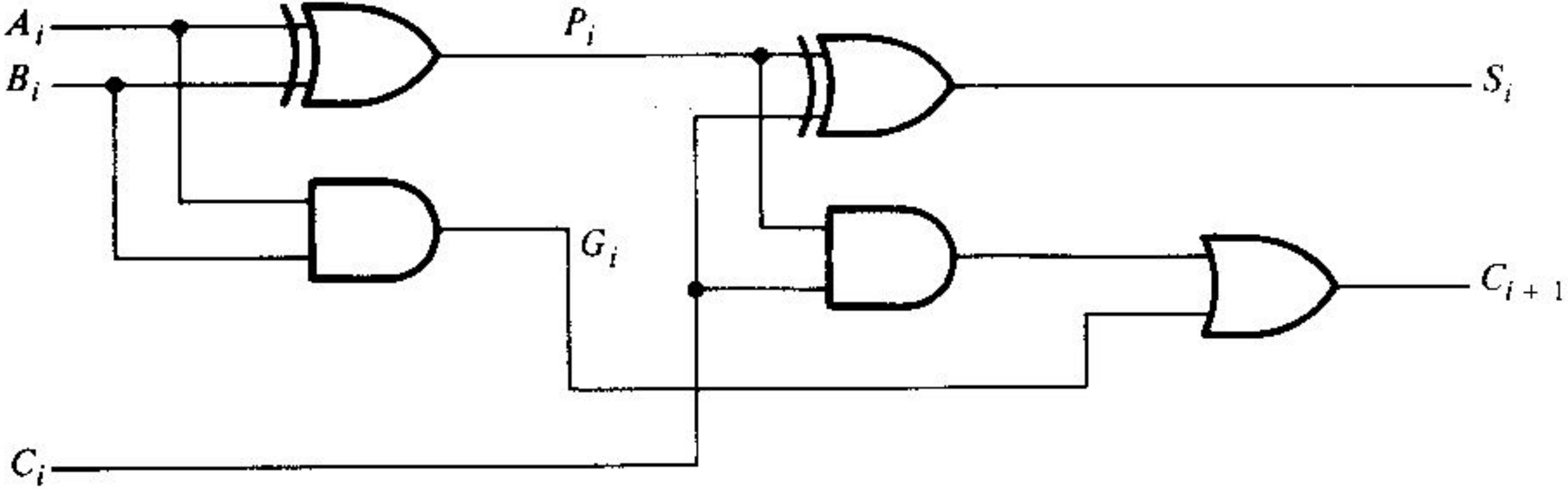
ADD/(SUB)'=1

- Gate 1,3,5,7 enabled and $C_0 = 0$
- B reaches gate 9,10,11,12
- Addition between A and B

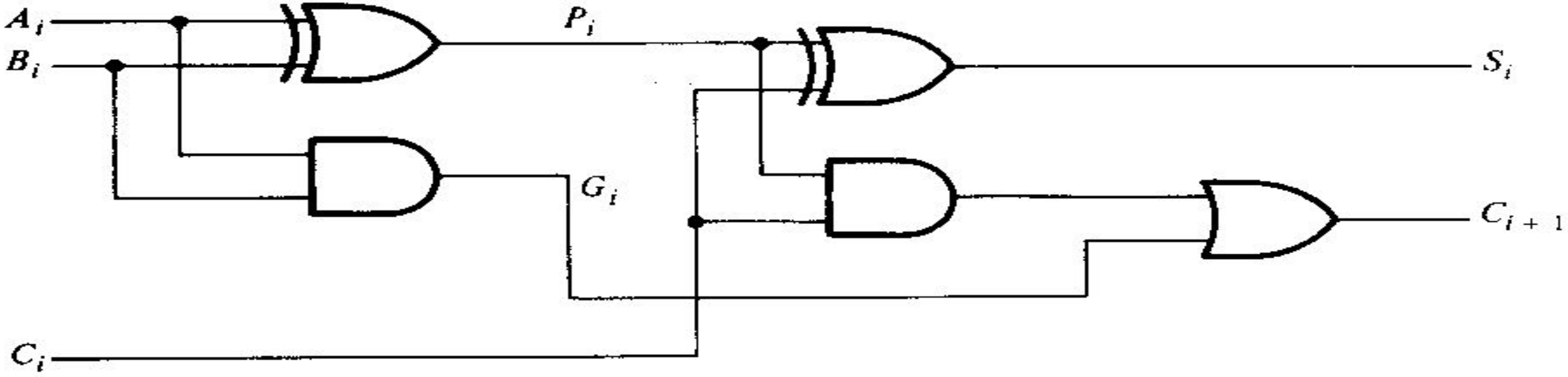
ADD/(SUB)'=0

- Gate 2,4,6,8 enabled and $C_0 = 1$
- B' reaches gate 9,10,11,12
- Addition between A and 2's Complement of B

Look Ahead Carry Adder



- C_i to C_{i+1}: propagate through two gate levels
- 4 FA: 2*4=8 gate levels
- N FA: 2n gate levels



$$P_i = A_i \oplus B_i$$

$$G_i = A_i B_i$$

$$S_i = P_i \oplus C_i$$

$$C_{i+1} = G_i + P_i C_i$$

Gi: Carry generate
Pi: Carry Propagate

$$P_i = A_i \oplus B_i$$

$$G_i = A_i B_i$$

$$S_i = P_i \oplus C_i$$

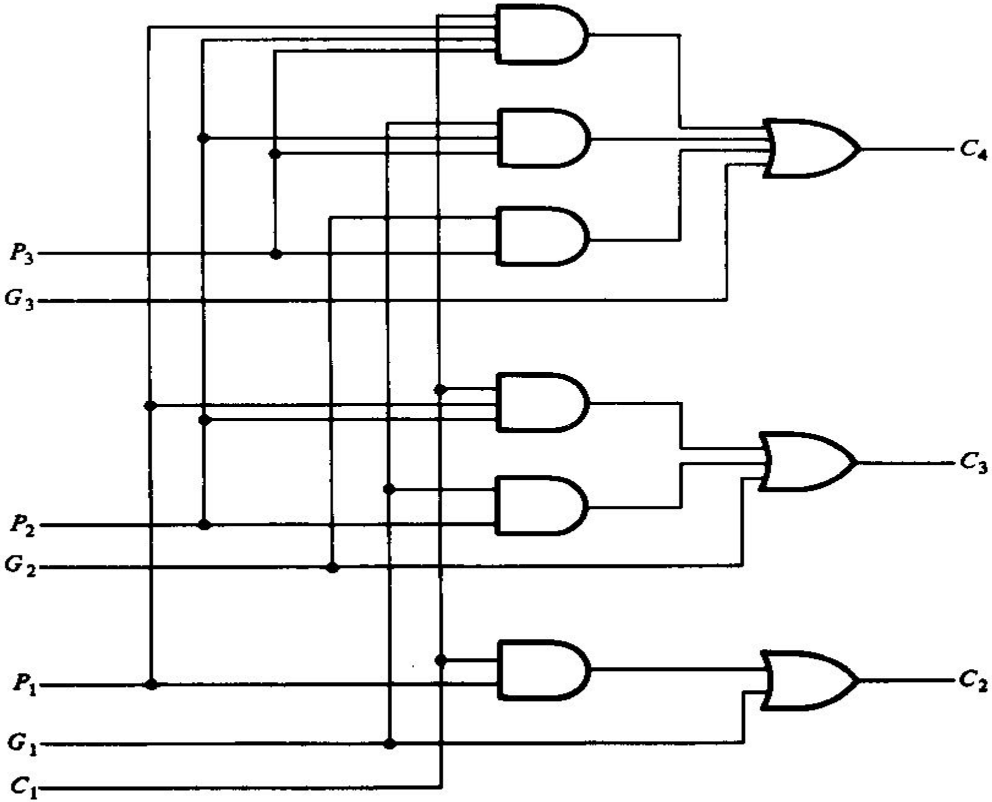
$$C_{i+1} = G_i + P_i C_i$$

$$C_2 = G_1 + P_1 C_1$$

$$C_3 = G_2 + P_2 C_2 = G_2 + P_2(G_1 + P_1 C_1) = G_2 + P_2 G_1 + P_2 P_1 C_1$$

$$C_4 = G_3 + P_3 C_3 = G_3 + P_3 G_2 + P_3 P_2 G_1 + P_3 P_2 P_1 C_1$$

LOOK AHEAD CARRY GENERATOR



$$C_2 = G_1 + P_1C_1$$

$$C_3 = G_2 + P_2C_2 = G_2 + P_2(G_1 + P_1C_1) = G_2 + P_2G_1 + P_2P_1C_1$$

$$C_4 = G_3 + P_3C_3 = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1C_1$$

Now C_4 doesn't have to wait for C_2 and C_3 to propagate

LOOK AHEAD CARRY ADDER